

ON-CHAIN ARCHITECTURE & ECONOMIC DESIGN

The Semesta Chronicle

Bringing a finished browser action-RPG on-chain on Rialo — what to put on the ledger, what to leave off it, how the assets should be issued, and what the economy must be able to survive.

PREPARED FOR · SEMESTA / ANAVELA UNIVERSE

TARGET NETWORK · RIALO — TESTNET LIVE AT PLAYGROUND.RIALO.IO

VERSION 1.0 · AUGUST 2026

BUILD FOR RIALO

Status of the game today: feature-complete and playable. 203 items, 57 weapons, 60 cosmetics, 8 classes, 20 monster species, 12 bosses, a 30-floor dungeon and a 50-tier seasonal pass — all of it live, and **none of it on-chain yet**.

How to read this document

Each section leads with its conclusion. If you read only the shaded box at the top of a section, you have the argument.

01	The case	<i>Why this game and this chain are an unusually close fit</i>
02	The boundary	<i>What goes on-chain, what must never go on-chain, and why</i>
03	On-chain structure	<i>Programs, state layout, and the Rialo primitives each one uses</i>
04	The asset design	<i>Three classes of NFT, and the test an asset must pass to become one</i>
05	The economy	<i>Two currencies, one firewall, and the sinks that already exist</i>
06	Sequencing	<i>Four releases, each shippable and each reversible</i>
07	Risk	<i>What kills this, ranked, with the control for each</i>
A	Appendix	<i>Open questions for Rialo engineering; primitive-to-feature map</i>

A note on what this document does not do

It does not argue that Semesta should be a financial product. The game is finished and it is fun without a token; that is the strongest possible position from which to add one, and the fastest way to lose it is to bolt on mechanics that make the game worse in order to make the asset tradeable. Every recommendation here is tested against one question: *does the game survive this change if the token goes to zero?* Where the answer is no, the recommendation is not made.

THE DISCIPLINE THIS DOCUMENT TRIES TO HOLD

Semesta's own in-game guide currently tells players, in plain language, that nothing is on-chain yet and that anyone asking them to connect a wallet today is not the team. That sentence was written deliberately. Nothing proposed here should be announced before it is true, and the guide should keep saying exactly what is and is not live at every stage.

Semesta already runs, off-chain, the exact machinery Rialo makes native

This is not a game looking for a chain. It is a game that has already built — in JavaScript, in a browser, against a wall clock — six workarounds that Rialo replaces with primitives.

THE ANSWER, FIRST

Go to Rialo, and go for the execution model rather than for the token. Semesta's distinguishing systems are all *asynchronous*: a house that finishes building in four hours whether or not the tab is open, a season that rolls over on a fixed calendar with no server told to do it, a world boss on a timer, crops that grow while you are elsewhere. On every other chain those are off-chain keepers, cron jobs and indexers — the “fragile tower” Rialo's own research names as the core problem. On Rialo they are conditional transactions and native workflows.

The token is the *last* reason to go, not the first. The first is that the game's architecture already has the right shape.

50ms

Rialo block time, parallel multi-concurrent-proposer consensus

0

Off-chain keepers a Semesta timer would need on Rialo

6

Existing game systems that map one-to-one onto a Rialo primitive

203

Items already defined, rarity-tiered and icon-complete

EXHIBIT 1

Every asynchronous system in Semesta is currently a workaround; each has a native Rialo counterpart

GAME SYSTEM, AS BUILT TODAY	THE WORKAROUND IT USES NOW	RIALO PRIMITIVE	WHAT CHANGES
Homestead construction cabin 20 min, cottage 4 h, villa 12 h	An absolute timestamp in <code>localStorage</code> . The client compares it to <code>Date.now()</code> on load. Trivially editable by the player.	Conditional transaction on a block-timestamp predicate	The build completes because the chain says so, not because a browser tab agreed
Season rollover 30-day pass, 4 seasons in rotation	Derived client-side from a fixed epoch precisely so that <i>no server is needed</i> . Every client computes the same boundary independently.	Time predicate evaluated in-block by every validator	The same design, but now the agreement is enforced rather than assumed
World boss scheduler every 180 s, shared in multiplayer	A server-side interval on a VPS behind a tunnel. If it dies, the boss stops spawning for everyone.	Recurring conditional transaction	A shared world event with no single machine responsible for it
Crop growth & daily reset 3 growth stages; midnight rollover	Wall-clock arithmetic on the client, re-checked once a second.	Time predicates; long-running workflow that sleeps and resumes	Growth continues across blocks without anything watching it
Account & three character slots Google sign-in, cloud-merged saves	Supabase row, custom merge logic. Getting the merge wrong once already risked deleting characters, so it is now a single tested pure function.	Rialo IPC — identity as a first-class capability; the Playground already mints a custodial testnet wallet from a Google sign-in	The exact sign-in the game already uses becomes the wallet. No seed phrase is ever shown to a player who does not want one
Deterministic world “the map is never sent”	Every client generates a byte-identical 200-cell world from fixed seeds; the multiplayer server sends only entities.	— already correct —	Nothing. This is the pattern on-chain games converge on, and it is already in place: state stays small because the world is a function of a seed

Sources: Semesta codebase, August 2026 — `systems/housing.js`, `systems/gamepass.js`, `server/`, `net/auth.js`, `world/terrain.js`. Rialo primitives per rialo.io Dev Portal and “Reactive Transactions” (April 2026).

Why the fit matters more than it sounds

Most games arriving on-chain are asked to become something they are not: to add a marketplace, a staking screen and a daily-login token in order to justify the ledger. Semesta needs none of that, because the parts of it that are hard to run — the timers, the shared clock, the season boundary, the account — are the parts Rialo is built to run. The chain is being asked to do infrastructure work, which is the only job it is unambiguously good at.

Put another way: the honest reason to deploy is that the game currently asks players to trust a VPS, a browser's clock and a Postgres row. Every one of those is a promise the team is making personally. On Rialo, three of them stop being promises.

The boundary is the design; almost every failed on-chain game drew it wrong

Nothing that runs at sixty frames per second goes on a ledger. What goes on the ledger is what must survive the studio.

THE ANSWER, FIRST

Put ownership, scarcity and the shared clock on-chain. Leave simulation, progression and all per-frame state off it. The test is not “can this be on-chain” — almost anything can. The test is “*does this need to outlive us, or be provable to a stranger?*” A player’s mythic weapon does. Their current stamina does not.

EXHIBIT 2

Draw the line at “must outlive the studio”, not at “could technically be written to state”

ON-CHAIN — MUST OUTLIVE US	OFF-CHAIN — WOULD ONLY MAKE IT WORSE
• Asset ownership. Which account holds which limited cosmetic, mythic weapon or pass. • Issuance & supply. How many of each limited piece exist, and the rule that mints them. • The season clock. Which season it is, when it ends, and the reward set bound to it. • Long timers. Homestead construction, prestige caches, crop maturity. • Achievement proof. First clears of the Hollow’s six bosses; Index category completion. • Identity. The account, its three character slots, and the link to a sign-in. • Token ledger. Balances, sinks, and any transfer.	• Everything per-frame. Position, facing, animation, camera, the 8 Hz position feed. • Combat resolution. Monster HP, aggro, damage rolls, the 20 species’ AI. • Hero progression. Level, XP, attribute points, skill trees. Not scarce, not tradeable, not interesting to a stranger. • The world itself. 200 cells of terrain generated from a seed. Storing it would be paying to keep a pure function. • Consumables and gold. Tonics, brews, forge stones, coins. Unlimited faucet by design. • Quest and story state. Personal, not scarce. • Cosmetic appearance data. The art is procedural and regenerates from an id.

THE RULE THAT DECIDES THE HARD CASES: IF A PLAYER WOULD BE ANGRY TO LOSE IT WHEN THE SERVERS CLOSE, IT GOES LEFT. IF THEY WOULD NOT NOTICE, IT GOES RIGHT.

Note the asymmetry in cost. Moving something left buys durability and pays in latency, fees and complexity. Moving something right buys speed and pays in trust. Semesta’s combat is 60 fps client-authoritative today; nothing in this document changes that.

THE FAILURE MODE TO AVOID, STATED PLAINLY

The common mistake is to put *progression* on-chain because it looks like ownership. It is not. Levels, skill points and quest flags are personal bookkeeping with no counterparty; putting them on a ledger adds a signature prompt to every kill and buys nothing. The games that did this are the ones players describe as “a wallet with a game attached”. Semesta’s own guide already draws this distinction correctly for pass XP versus hero XP; the same instinct applies here.

What the boundary implies for the client

Semesta’s renderer, physics and combat stay exactly as they are. The on-chain layer sits beside them and is consulted at four moments only: when an asset is *granted*, when an asset is *spent or burned*, when a *timer* is started or read, and when the *season* turns. Everything else in the game never touches it. That is a small enough surface that it can be added without rewriting the game, and — equally important — removed again if the chain is the wrong bet.

Five programs, not one; the state stays small because the world is a seed

The whole on-chain footprint of a 200-cell open world with 203 items fits in five programs, because none of them stores a world.

THE ANSWER, FIRST

Split by lifetime, not by feature. Registry and Ledger change rarely and must never be lost. Season and Timers churn every block. Identity is read constantly and written almost never. Bundling them produces one program that is simultaneously hot, precious and unupgradeable — the shape that forces teams into risky migrations later.

Programs are separated by how often they change, so the precious state is never in the hot path

PROGRAM	OWNS	RIALO PRIMITIVE USED	WRITE FREQUENCY	IF IT FAILS
Identity	Account, its three character slots, the link from a sign-in to a wallet, 2FA policy	Rialo IPC; Omni Account	Once per account, then almost never	Nobody can log in. Highest blast radius, lowest change rate — so it ships first and then stops moving
Registry	The item catalogue: 203 ids, rarity, slot, and the supply rule for anything limited	Plain state; upgraded via Gauss-style protocol upgrade discipline	Once per content release	No new items can be defined. Existing holdings unaffected
Ledger	Who owns which limited asset; transfers; burns; the soulbound flag	Plain state; Hybrid Concurrency Control for parallel writes	On grant, trade or burn only	Assets freeze in place. Nothing is lost; the game keeps running off-chain
Season	Current season index, its reward track hash, the pass a player holds, prestige counters	Conditional transaction on a time predicate	Per player action; one global rollover per 30 days	The season stops turning. Players keep their claimed assets
Timers	Homestead builds, crop maturity, prestige caches, the world-boss schedule	Conditional transactions; long-running workflows that sleep and resume	Continuous, but only on triggers	Timers stall. The single most recoverable failure — and the one that today depends on a browser tab

Rialo VM is RISC-V with SVM compatibility today and EVM/MoveVM signalled to follow, so the implementation language is a decision that can be deferred past the architecture. That is a genuine advantage: the split above does not depend on which one is chosen.

The state-size argument, quantified

The reason this is affordable is that Semesta already refuses to store its world. Terrain, decor, camps, chests and landmark placement are all derived from fixed seeds, which is why the multiplayer server can be “one small box” and why the map is never sent over the wire. The same property carries on-chain: the ledger stores *which account owns item id X*, never what item X looks like, where it is, or what the ground beneath it is made of. The art regenerates from the id on the client, procedurally, in a 16-pixel canvas.

Because the art is procedural and the id is the asset, a Semesta NFT does not need image hosting, IPFS pinning, or a metadata gateway — the three things that quietly break most NFT collections within a few years. The token points at an id in the Registry; any client that can run the game can draw it. This is an accident of the original “never add asset files” house rule, and it is worth a great deal.

Reactive transactions, in the game's own vocabulary

EXHIBIT 4

A homestead build is a predicate; nothing watches it and nothing needs to be online



The same four steps describe the season rollover, a prestige cache, a crop reaching stage three, and the 180-second world boss. One primitive, four features — which is what Rialo's research calls supermodularity, and is the practical reason to prefer it here.

Three classes of asset, and a four-question test before anything becomes one

Semesta has 203 items. Roughly twenty-three of them should ever be tokens. The discipline is in the refusals.

THE ANSWER, FIRST

Tokenise proof and scarcity; refuse to tokenise inventory. Three classes: **SOULBOUND** for things that are proof of what a player did, **TRADEABLE** for things that are genuinely scarce and genuinely someone else's to want, and **OFF-CHAIN** for everything else — which is the large majority, and should stay that way.

THE ON-CHAIN TEST

An item becomes an NFT only if it passes all four. Any one failure and it stays an ordinary game item.

1	Is supply genuinely limited?	Not “rare drop”. Limited by a rule that can be stated in one sentence and checked by a stranger. Semesta already tags three such sets: Lanterns (8), The Hollow (9), Keeper's Vault (6).
2	Would anyone want it who did not earn it?	A cosmetic passes. A stack of forge stones does not — it is a commodity with an unlimited faucet, and tokenising commodities is how game economies inflate.
3	Does trading it damage the game?	If an item is the reward for a fight, and it can be bought instead, the fight is now optional. This is the question that sends most weapons to the off-chain column.
4	Does it still make sense if the market is dead?	The asset must be worth owning for what it does in the game, with zero bid. If the only reason to want it is resale, it is not a game asset.

Applying the test to Semesta's existing content sends most of it off-chain — which is the correct result

ASSET	COUNT	CLASS	REASONING
SOULBOUND — PROOF OF WHAT A PLAYER DID. NEVER TRANSFERABLE.			
Genesis Lanternkeeper	1 / account	SOULBOUND	The account itself, minted free on first sign-in during the testnet period. It is an identity, not an asset — it carries the three character slots and the player's history. Making it tradeable would make accounts a commodity and invite exactly the farming the game does not need.
Hollow first-clear seals	6	SOULBOUND	One per lord defeated, per difficulty. These are the only durable record that a player cleared floor 30 on hard. Worthless to a buyer by construction, which is the point: it is a claim about you.
Index completion marks	7	SOULBOUND	One per collection category completed. The Index already tracks 200 entries and already pays out on completion; this adds a permanent record and no new mechanic.
TRADEABLE — GENUINELY SCARCE, AND LOSING THE FIGHT TO OWN IT IS FAIR.			
The Keeper's Vault set	6 per season	TRADEABLE	The strongest candidate in the game, and it already exists. The Vault issues each piece exactly once per player, never repeats, and is reachable only through a currency that cannot be bought. Supply per season is therefore a known function of how many players finished the track — a mint rule you can state in a sentence.
Season passes	2 types	TRADEABLE	Keeper's Seal and Keeper's Sigil are already <i>items bought in a shop and consumed</i> , not a button in a menu. That is precisely a bearer instrument: transferable while sealed, burned on activation. The design work is done.
Mythic weapons	~9	TRADEABLE	Only with the forge history attached, so a +9 blade carries the record of the stones burned into it. Without that provenance they are just stats and fail test 3.
OFF-CHAIN — THE LARGE MAJORITY. LEAVING THESE ALONE IS A DECISION, NOT AN OMISSION.			
Ordinary cosmetics	37	OFF-CHAIN	Unlimited supply from the gacha. Fails test 1.
Weapons, tiers 0–3	48	OFF-CHAIN	Craftable on demand from common materials. Fails test 1 and 3.
Pets & mounts	25	OFF-CHAIN	Companions are quest and gacha rewards; trading them turns a collection into a shopping list. Fails test 3. Revisit only if a genuinely limited companion is ever made.
Consumables, gold, materials	~80	OFF-CHAIN	Unlimited faucets by design. Tokenising them is the classic inflation error.
The homestead	1 / player	OFF-CHAIN	Deliberately not land. Semesta has exactly one buildable site per player, on purpose — the four-parcel version was cut because identical plots are a choice with no content in them. Re-introducing scarce land to sell it would undo a design decision that was made for the game's benefit.

Counts are from the live item table, August 2026. The tradeable column totals roughly 23 assets against 203 items — about 11%. If that ratio rises much, it is a sign the test is being applied loosely.

WHY THE VAULT IS THE RIGHT FIRST COLLECTION

Most NFT drops have to invent a scarcity story. The Vault's is already true and already shipped: six pieces, no duplicates ever, a currency that has no price because it cannot be bought, and a set that completes in exactly six tokens. A player who owns all six this season owns proof of a finished season. Season two rotates the set, which makes season one's set permanently closed — without anyone having to promise it will be.

Two currencies with a firewall between them, and sinks that already outnumber faucets

The single most common cause of death in GameFi is an unlimited faucet wired to a market. Semesta's protection is to never connect them.

THE ANSWER, FIRST

Gold stays soft, infinite and off-chain. \$SEMESTA stays hard, capped and on-chain. There is no exchange rate between them, in either direction, ever. Gold buys consumables, farm plots, boat keys and season passes. \$SEMESTA does exactly three things: it mints limited assets, it settles trades between players, and it pays for seasonal entry. A player who never touches it can still finish the pass, clear the Hollow and complete the Index.

EXHIBIT 6

The firewall is the whole design: grinding produces gold, and gold can never become the token

GOLD — SOFT CURRENCY, UNCHANGED	\$SEMESTA — HARD CURRENCY, CAPPED
Faucets: monster drops, fishing, farming, chests, quests, the pass track, daily check-in, level rewards. Effectively unlimited. Sinks that already exist: gacha (100c, or 900c for ten), forging +1...+9 with failure, seeds, tonics, brews at 220–260c, farm plots, the 480c Donut Boat key, house tiers at 200c and 600c, and both season passes at 1,000c and 3,000c. Status: a working closed loop. It needs no chain and gets none.	Faucets: seasonal emission only, distributed against the pass track and Hollow clears. Emission per season is fixed in advance and declines on a published schedule. Sinks: minting a Vault piece, trading fees, seasonal entry for the on-chain track, and burning on pass activation. Status: does not exist yet. Nothing below should be announced as if it does.

NO SWAP. NO CONVERSION. NO “GOLD TO TOKEN” RATE AT ANY POINT — THAT SINGLE LINK IS WHAT TURNS A GAME ECONOMY INTO A FARM.

The asymmetry is deliberate and worth defending under pressure: gold is allowed to inflate because it buys only things the game prints. The token is not allowed to inflate because it buys things the game cannot reprint.

The Keeper's Token is already the right bridge, and it was not designed to be

The pass currently pays a currency — the Keeper's Token — that *cannot be purchased at any price*. Its only sources are the reward track (2 on the free row, 5 with Basic, 10 with Premium) and the endless prestige caches past tier 50. It buys only Vault pieces, and the Vault never repeats.

That is, structurally, a mint pass. It has a supply that is a function of play, a sink that consumes it, and a distribution curve that already separates free players from paying ones without locking anyone out. The on-chain version should change nothing about how it is earned; it should only make the resulting asset real.

Season economics, computed from the live track rather than assumed

QUANTITY	FREE	BASIC	PREMIUM	SOURCE
Pass XP to finish the track	32,950	32,950	32,950	50 tiers, 120 → 1,198 per tier
Approx. hours of play to finish	~9.2	~8.4	~7.4	at ~60 XP/min, before the pass XP perk
Keeper's Tokens earned	2	5	10	track milestones
... plus prestige caches	1 / 1,500 XP	1 / 1,500 XP	1 / 1,500 XP	endless past tier 50
Vault pieces obtainable	2 of 6	5 of 6	6 of 6	no duplicates, ever
Seasons to complete the set, free	3	—	—	and the set rotates each season
Gold cost of the pass	0	1,000	3,000	bought as an item at Pip's Shop

The important line is the last but one. A free player can complete a Vault set, but it takes three seasons and by then the set has rotated — so paying buys *time and completeness*, never power. No number in this table changes a combat outcome.

WHERE THE PUBLISHED ROADMAP NEEDS REVISITING

The in-game roadmap currently lists “**Hold to Play — minimum 1,000 SEMESTA**” as phase two. This should be cut. It contradicts the game's own strongest design principle — the free reward row exists specifically so that nobody is locked out — and it converts the top of the funnel into a purchase decision made before the player has seen anything. A holding gate also produces the worst possible first impression in a market already suspicious of exactly this pattern.

If holders need a benefit, give them one that is additive rather than exclusionary: a cosmetic frame, priority in a queue, a seasonal title. The difference between “holders get something extra” and “non-holders get nothing” is the difference between a game with a token and a token with a game.

Four releases, each of which is worth shipping even if the next one never happens

The testnet is live now. The correct first release contains no token and no NFT.

THE ANSWER, FIRST

Ship identity first, assets second, market third, token last. Each release is independently valuable, independently reversible, and does not require believing the next one will work. A team that ships the token first has spent its credibility before it has proved it can run a chain integration at all.

Sequencing by reversibility: the earliest releases are the ones that can be undone quietly

#	RELEASE	SHIPS	PROVES	IF IT GOES WRONG
R1	Keeper's Passport testnet, now	Google sign-in mints a custodial Rialo wallet. The account and its three character slots move from Supabase to the Identity program. A free soulbound Genesis Lanternkeeper on first login. <i>No token, no NFT anyone can sell, no wallet UI.</i>	That the game can read and write chain state at all, and that a non-crypto player can be onboarded without ever seeing a seed phrase	Fall back to Supabase. The player never knew. This is the entire reason it goes first
R2	The Vault, on-chain testnet	The six Vault pieces are minted on-chain when the Keeper's Token is spent. Season index, prestige counters and the pass move to the Season program. Homestead timers become conditional transactions. Still non-transferable	That issuance is correct under load, and that the season boundary agrees across every client without a server	Assets stay minted; trading was never enabled, so nobody is holding a position. Revert the client to local timers
R3	Transfer testnet → mainnet	Vault pieces, passes and mythic weapons become transferable. Player-to-player trade. Soulbound classes stay soulbound. A published supply page for every limited set	That a real market forms and that it does not distort how the game is played	Trading can be paused. The assets remain owned; only the transfer function is gated
R4	\$SEMESTA mainnet	The token, its emission schedule, and the sinks in Exhibit 6. Genesis Lanternkeeper holders from R1 are recognised. Gacha optionally accepts \$SEMESTA <i>alongside</i> gold, never instead of it	Nothing new technically — by this point every mechanism has run for months. That is the point of putting it last	This is the release that cannot be quietly undone, which is exactly why it comes after three that can

R1 and R2 are the releases that need Rialo specifically. R3 and R4 are ordinary once R1 and R2 work. A useful test of the plan: if Rialo's mainnet slipped a year, R1 and R2 would still be the right work to do on testnet.

What can start this week

The Playground already demonstrates the exact onboarding Semesta needs — sign in with Google, receive a custodial testnet wallet, hold a points balance that persists across apps in a shared catalogue. Semesta's account layer is a near-identical shape and already routes every entry through a sign-in. R1 is therefore not a research project; it is a substitution.

The second thing that can start immediately is a supply page. Before any asset is tradeable, publish how many of each limited piece exist and the rule that mints them. Doing this before there is a market is what separates a collection from a promise.

What actually kills this, ranked, and what to do about each

Two of the top three risks are not technical, and none of the top three is “the chain goes down”.

THE ANSWER, FIRST

The largest risk is that the token makes the game worse. Semesta is finished and enjoyable now; the failure mode is not a hack, it is a series of small concessions that turn a game people play into a yield product people leave. Every control below is a way of noticing that early.

The risk register is dominated by design and regulatory exposure, not by infrastructure

#	RISK	SEVERITY	WHY IT HAPPENS	CONTROL
1	The economy eats the game	CRITICAL	Systems are tuned for token velocity rather than for play. Grind loops get longer because longer grinds emit more. The gacha becomes the main screen.	The firewall in Exhibit 6, and a standing rule: no combat, drop or progression number may be changed for a token reason. Publish the tuning diffs each season
2	Gacha + tradeable assets reads as gambling	CRITICAL	A paid random draw whose outputs have a cash market is regulated as gambling in a growing number of jurisdictions — and Indonesia's position on both gambling and crypto is strict and moving.	Keep the gacha on <i>gold only</i> . The Vault is deterministic by construction — no duplicates, six tokens for six pieces — so the on-chain path contains no random paid draw at all. Take local counsel before R3, not after
3	Tokens or NFTs treated as securities	HIGH	Marketing that implies appreciation, a team allocation with a vesting narrative, or utility described as yield.	Describe assets by what they do in the game and nothing else. No projections, no APY language, no “floor” talk from the team's own accounts
4	Client-authoritative progression is farmed	HIGH	Combat, XP and drops are resolved on the client today. If pass XP mints tokens, forging the client mints tokens.	Emission must key off <i>server- or chain-observed</i> events only. This is real work and it is the main engineering cost of R4 — budget it honestly rather than discovering it late

5	Sybil farming of free rewards	MEDIUM	Free tokens per account plus cheap accounts equals a farm.	Genesis Lanternkeeper is soulbound and minted once per verified sign-in; emission weights by account age and by Hollow progress, which cannot be bottled cheaply
6	Custody and key loss	MEDIUM	Players who signed in with Google do not think of themselves as holding keys.	Custodial by default, as the Playground already does. Self-custody is an opt-in for players who ask. Rialo's 2FA and social-recovery primitives are the reason this is tenable
7	Chain immaturity	MEDIUM	Rialo is young. Testnet is live; mainnet timing is not in the team's control.	R1–R2 are useful on testnet alone. Keep the off-chain path in the codebase behind a flag until R3 — the game must remain fully playable with the chain switched off
8	Asset permanence claims	LOW	Most collections break when a metadata host lapses.	Structurally avoided: the art is procedural and regenerates from the id. Worth stating publicly, because it is a real differentiator

Risks 1 and 2 are the ones to re-read before every release. They are also the two that no amount of engineering will fix after the fact.

A TEST TO APPLY BEFORE EACH RELEASE

Show the build to somebody who does not own crypto and does not intend to. If their experience of the game got worse in any way — an extra click, a wallet prompt, a screen they must dismiss, a reward they can no longer reach — the release is not ready, regardless of what it does for holders. Semesta's audience is players first; the guide, the prologue and the free reward row all already assume that.

Open questions for Rialo engineering

These are the answers that would change the architecture above. They are worth asking before R1, not during it.

#	QUESTION	WHY IT CHANGES THE DESIGN
1	What is the cost and latency ceiling of a conditional transaction that sleeps for 12 hours ?	The villa build is 12 h. If long-lived predicates are priced per block evaluated rather than per trigger, the timer design changes shape entirely
2	How many predicates can one account hold concurrently?	A busy player has a house timer, up to 8 crop timers and a prestige counter at once — roughly 10 per account, times the player base
3	Is there a per-block cap on triggered transactions, and what happens on the season boundary when <i>every</i> player's season predicate fires in the same block?	This is the one genuinely adversarial moment in the design: a synchronised global trigger. If it needs smoothing, the season rollover must be staggered per account
4	What are the guarantees and the rate limit on Rialo Edge HTTPS calls from a contract?	Determines whether the multiplayer server can be retired or must remain
5	Does Rialo IPC expose the Playground's Google-to-custodial-wallet flow as a reusable primitive, or is that Playground application code?	R1 is a substitution if the former and a build if the latter. This is the single highest-value answer on this page
6	Is on-chain randomness available to contracts, and is it manipulable by a proposer?	The gacha stays off-chain in this plan, so this is not blocking — but it determines whether it ever could move
7	What is the migration story for the Registry as items are added — does Gauss-style upgrade apply to application programs or only to the protocol?	Semesta ships new items regularly. A catalogue that cannot grow is a catalogue that forces a redeploy
8	What does Rialo Cruise gasless transaction sponsorship cost the sponsoring application?	Players must never see a gas prompt for claiming a pass tier. If sponsorship is affordable, the UX problem disappears
9	Is there a testnet reset schedule, and what survives one?	Genesis Lanternkeepers minted in R1 are a promise to early players. If testnet state is wiped, that promise must be honoured off-chain and re-minted

Appendix B — primitive to feature map

Read this as the short version of the whole document: the left column is what Rialo publishes, the right is the Semesta feature that consumes it.

RIALO COMPONENT	PUBLISHED CAPABILITY	SEMESTA FEATURE THAT USES IT
Execution Engine	Event-driven execution with conditional transactions	House builds, crop growth, prestige caches, world-boss timer
Workflow	Native automation; workflows that sleep and resume across blocks	The 30-day season lifecycle, from rollover to reward reset
IPC	Identity, privacy and compliance as first-class capabilities	Google sign-in → account → three character slots; 2FA; recovery
Omni Account	One account transacting across networks	One Keeper's Passport, three heroes, any client
Read Path	State read directly from validators, ~100 ms, no indexer	Reading owned assets at world-build time without running infrastructure
Edge	Bidirectional Web2 interactivity without middleware	Possible retirement of the multiplayer VPS; leaderboards
Cruise	Native gasless transactions	Claiming a pass tier without a fee prompt

Consensus	Parallel multi-concurrent proposers, 50 ms blocks	Trades and mints that settle inside a session rather than between them
VM	RISC-V; SVM-compatible, EVM/MoveVM signalled	Language choice can be deferred past the architecture

Sources and status. Rialo capabilities are as published on rialo.io (Dev Portal; “Introducing Rialo”, September 2025; “Reactive Transactions: A Model for Native Automation on Rialo”, April 2026) and as observed on the live Playground at playground.rialo.io, August 2026, where the testnet reported *online* and two applications were listed. Performance figures quoted are Rialo's own published claims and have not been independently benchmarked here; the 50 ms block time is taken from the Dev Portal's Consensus description. Semesta figures are measured directly from the codebase at the same date: 203 items, 57 weapons, 60 cosmetics across 6 rarities, 3 limited sets (8 / 9 / 6 pieces), 17 pets, 8 mounts, 20 monster species, 12 bosses, 8 classes, a 30-floor dungeon at 3 difficulties, and a 50-tier seasonal pass totalling 32,950 pass XP.

Nothing described in this document is live on-chain. The game is playable and complete off-chain today, and its in-game guide correctly tells players so.